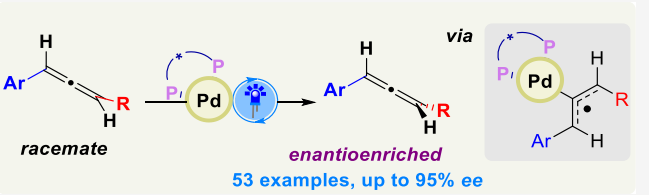
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Photoexcited Palladium-Catalyzed Deracemization of Allenes

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ABSTRACT: The different enantiomers of specific chiral molecules frequently exhibit disparate biological, physiological, or pharmacological properties. Therefore, the efficient synthesis of single enantiomers is of particular importance not only to the pharmaceutical sector but also to other industrial sectors, such as agrochemical and fine chemical industries. Deracemization, a process during which a racemic mixture is converted into a nonracemic product with 100% atom economy and theoretical yield, is the most straightforward method to access enantioenriched molecules but a challenging task due to a decrease in entropy and microscopic reversibility. Axially chiral allenes bear a distinctive structure of two orthogonal cumulative π-systems and are acknowledged as synthetically versatile synthons in organic synthesis. The selective creation of axially chiral allenes with high optical purity under mild reaction conditions has always been a very popular and hot topic in organic synthesis but remains challenging. Herein, a photoexcited palladium-catalyzed deracemization of nonprefunctionalized disubstituted allenes is disclosed. This method provides an efficient and economical strategy to accommodate a broad scope of allenes with good enantioselectivities and yields (53 examples, up to 96% yield and 95% ee). The use of a suitable chiral palladium complex with visible light irradiation is an essential factor in achieving this transformation. A metal-to-ligand charge transfer mechanism was proposed based on control experiments and density functional theory calculations. Quantum mechanical studies implicate dual modes of asymmetric induction behind our new protocol: (1) sterically controlled stereoselective binding of one allene enantiomer under the ground-state and (2) facile, noncovalent interaction-driven excited-state isomerization toward the opposite enantiomer. The success of this newly established photochemical deracemization strategy should provide inspiration for expansion to other multisubstituted allenes and will open up a new mode for enantioselective excited-state palladium catalysis.



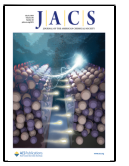
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INTRODUCTION

The different enantiomers of specific chiral molecules, such as small-molecule drugs, frequently exhibit disparate biological, physiological, or pharmacological properties.[1] It is not surprising that one enantiomer of a chiral drug is clinically effective and the other optical isomer can produce unwanted or even catastrophic side effects.[2,3] Therefore, the efficient synthesis of single enantiomers is of particular importance not only to the pharmaceutical sector but also to other industrial sectors, such as agrochemical and fine chemical industries.[4] Mother Nature produces sugars and amino acids as a single enantiomer in biological systems, which results in the homochirality of living organisms.[5] Regrettably, the range of chiral pools provided by nature is limited in terms of structure and stereochemistry. Asymmetric synthesis has been evolved as a powerful synthetic platform for the construction of diverse enantioenriched molecules in the most efficient and practical manner during the past few decades.[6,7] However, the identification of a highly enantioselective chiral reagent[8] or catalyst[9] for a certain reaction is not always an easy task. A well-established alternative strategy relies on classic or kinetic resolution process of a racemic mixture but with a maximum 50% theoretical yield of the desired product and the recovered substrate.[10−12]

Deracemization, as defined by Pirkle,[13] is a process "in which a racemate is made nonracemic by increasing the quantity of one enantiomer at the expense of the other". The deracemization process offers a 100% atom-economic protocol without the release of any byproducts. However, since such a process represents a reaction in the energetic uphill direction, external energy is required to provide the entropy balance. Moreover, the principle of microscopic reversibility makes this process kinetically challenging (Figure 1a).[14] Visible (solar) light, as a sustainable energy source, can supply the necessary external perturbation to drive the deracemization reaction forward. Therefore, photochemical deracemization mediated by visible light has gained attention from the chemistry community, and some breakthroughs have been achieved recently.[15−32] Bach and co-workers reported visible-light-driven deracemization of amide-containing allenes mediated by the hydrogen bonding between chiral photosensitizer and

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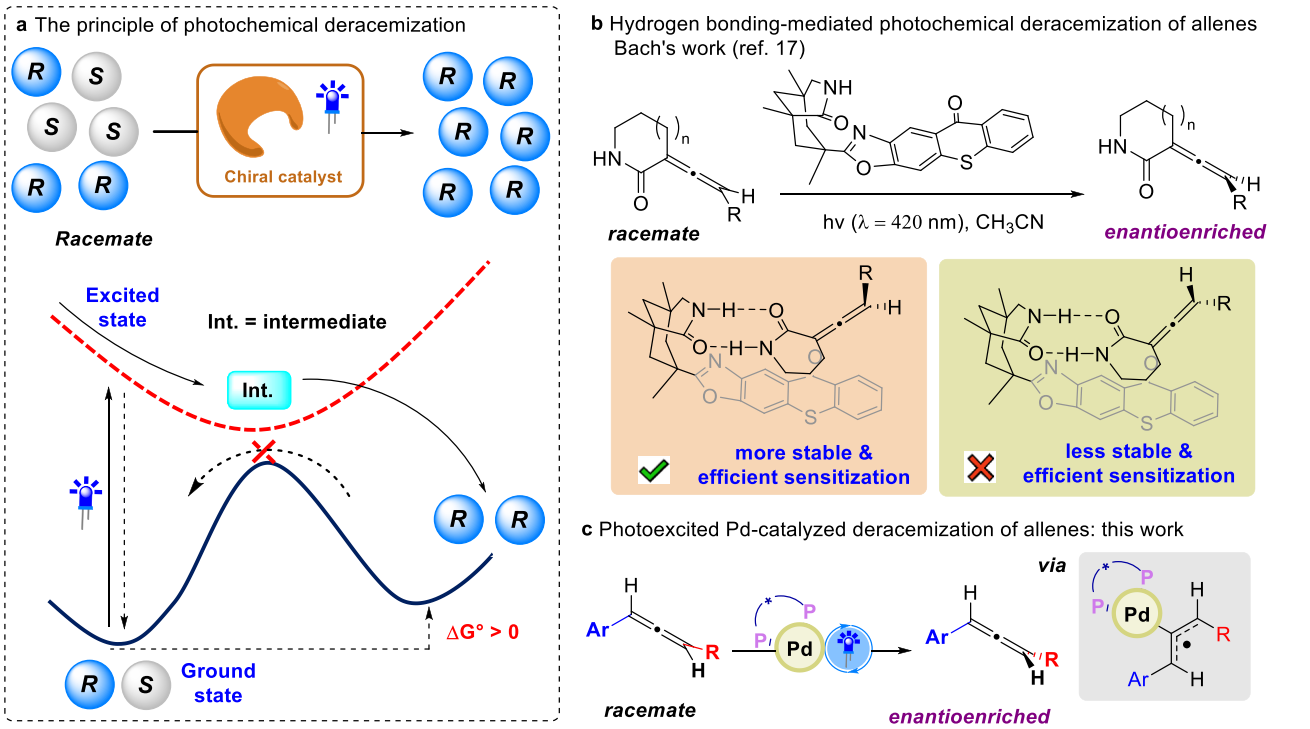


Figure 1. Photochemical deracemization of allenes.

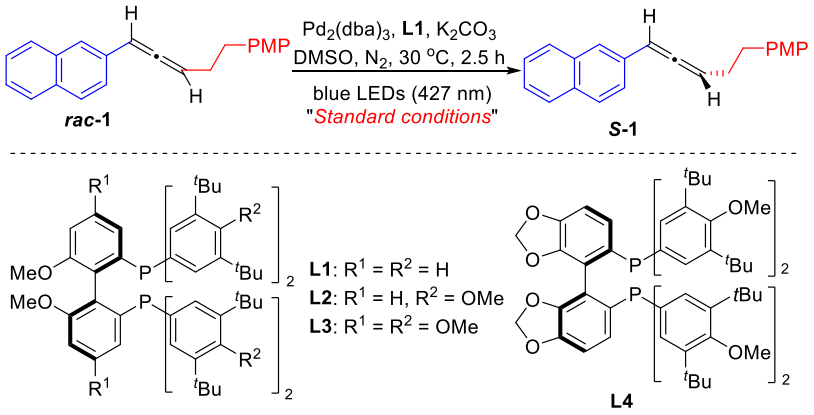
substrates (Figure 1b).[17] The different overall sensitization rates of two enantiomers are responsible for this deracemization. Knowles/Miller[18] and Bach[23] also demonstrated photochemical deracemization of 5-substituted hydantoins and 4-substituted imidazolidin-2-ones, respectively, through asymmetric hydrogen atom transfer. Meggers/Chen[21] and Luo[24] achieved deracemization of α-branched ketones and aldehydes, respectively, through photochemical asymmetric proton transfer. These achievements showcase that the photochemical processes are valuable synthetic means for the deracemization. However, racemic substrates must be sophisticatedly designed and prefunctionalized properly.

Axially chiral allenes are acknowledged as synthetically versatile synthons in organic synthesis and bear a distinctive structure of two orthogonal cumulative π-systems.[33−35] Moreover, chiral allenes are ubiquitous in advanced functional materials[36] and naturally occurring products.[37,38] Undoubtedly, the exploration of high-efficiency approaches for asymmetric synthesis of optically active allenes has received considerable attention of chemists.[39−42] Although a series of efficient approaches, involving allenyl cations,[43] allenyl anions,[44−46] allenyl radicals,[47,48] and molecular rearrangements,[49] were established to synthesize enantioenriched allenes, efficient protocols toward this goal with a concise manipulation are still highly sought after. Given that the synthesis of racemic allenes are well established,[50,51] access to optically active allenes through deracemization of racemic allenes is a highly attractive but challenging strategy and deserves to be investigated. The initial work by Weiss and coworkers demonstrated that photochemical deracemization of penta-2,3-diene with a chiral sensitizer through an energy transfer pathway was feasible, but with very small enantioselectivity (ca. 3.4% ee).[52] Extraordinary works by Bach and coworkers elegantly showed that if a hydrogen bonding between chiral sensitizer and the allenes was introduced, the efficiency of the photocatalytic deracemization of allenes was improved dramatically (up to 97% ee).[17] The different association constants and sensitization efficiencies of hydrogen bonding complexes between the chiral thioxanthone and each enantiomer of the allene racemate ensure the observed deracemization (Figure 1b).

Transition metal catalysis at its photoexcited states has attracted significant attention for the discovery of unprecedented transformations.[53−56] In contrast to metallaphotoredox catalysis, which employs photosensitizers and transition metals synergistically,[57−61] photoexcited transition metal catalysis uses a single transition metal complex to harvest light and enable bond forming or breaking events. However, the double duty of transition metal catalysis makes the asymmetric version of this catalytic mode formidably challenging, and few enantioselective excited transition metal catalytic transformations were reported.[62−69] Very recently, our group succeeded in achieving the kinetic resolution of 2-styrylpyrrolidines enabled by photoexcited chiral copper complex-mediated alkene $E \rightarrow Z$ isomerization.[70] The different overall sensitization rates between the chiral copper complex and the two enantiomers of 2-styrylpyrrolidines led to excellent kinetic resolution efficiency. This success inspires us to investigate photoexcited transition metal-mediated deracemization of racemic allenes (Figure 1c). Energy from visible light can compensate for the entropy loss. The complexes of a transition metal, a chiral ligand, and chiral allenes are diastereomeric and chemically distinguishable, wherein the principle of microscopic reversibility is not applicable. Here, we show that fine-tuning of ligand−substrate interactions can precisely create thermodynamic biases to encompass the rapid photochemical deracemization of allene racemates. The transition metal binds to the allene directly, and the preinstalled binding site can be avoided, which can expand the scope of allenes significantly.

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Table 1. Investigation on the Photochemical Deracemization Condition Parameters[a]



entry	variation from standard conditions	yield/%[b]	ee/%[c]
1	none	>99 (97)	94
2	**L2** instead of **L1**	>99 (95)	89
3	**L3** instead of **L1**	>99 (85)	70
4	**L4** instead of **L1**	>99 (80)	59
5	CH_3CN instead of DMSO	97 (55)	18
6	DMA instead of DMSO	91 (77)	70
7	Cs_2CO_3 instead of K_2CO_3	>99 (95)	90
8	KOAc instead of K_2CO_3	>99 (74)	47
9	456 nm LEDs instead of 427 nm LEDs	>99 (88)	76
10	$PdCl_2$ instead of $Pd_2(dba)_3$	>99 (50)	0
11	without light	>99 (50)	0
12	without ligand	>99 (50)	0
13	without $Pd_2(dba)_3$	>99 (50)	0
14	without K_2CO_3	>99 (69)	38

[a]Reactions were carried out with a solution of *rac*-**1** (0.1 mmol), $Pd_2(dba)_3$ (2.5 mol %), **L1** (6 mol %), and K_2CO_3 (0.15 mmol) in DMSO (2.0 mL) at 30 °C under 40 W blue LEDs (427 nm) for 2.5 h. [b]The overall yield of *S*-1 and *R*-1 was determined by GC with an internal standard (tetradecane). The yield of *S*-1 was calculated based on the GC overall yield and enantiomeric excess (ee) values and is given in the parentheses. [c]Enantiomeric excess (ee) values were determined by HPLC on a chiral stationary phase. PMP = *p*-methoxyphenyl.

RESULTS AND DISCUSSION

Reaction Investigation. Toward this end, we initiated the investigation utilizing configurationally stable aryl-alkyl disubstituted allene *rac*-**1** as a model substrate. After a quick screening, it was regretfully found that the chiral copper complex under visible light irradiation was an ineffective catalytic system for the deracemization of allene *rac*-**1**. We had to turn our attention to other transition metal-based photocatalytic systems. Given that allenes can coordinate to palladium, we focused on the development of photoexcited palladium catalysis for the deracemization of allenes. This adjustment turned out to be fruitful. After careful and thorough investigation on the reaction parameters (for more details, see Tables S1−S5 in the Supporting Information), the photochemical deracemization of allene *rac*-**1** could be achieved in the presence of a combination of $Pd_2(dba)_3$ (2.5 mol %)/diphosphine ligand **L1** (6 mol %) and K_2CO_3 as a base in a solution of dimethyl sulfoxide (DMSO) under irradiation with blue light-emitting diodes (LEDs) (λ = 427 nm) at room temperature. The photostationary state was reached in 2 h, and the desired enantioenriched allene S-1 was obtained in an almost quantitative yield (>99% GC overall yield) and excellent enantioselectivity (94% ee) (Table 1, entry 1). Other ligands (entries 2−4), solvents (entries 5 and 6), and bases (entries 7 and 8) were also explored, but no better results were obtained. The investigation on visible-light sources indicated that the emission wavelength of LEDs was crucial to this photochemical process, and the longer wavelength led to lower deracemization efficiency (entry 9). Palladium(II) complexes were ineffective in this reaction (entry 10). Control experiments elucidated that visible light, the ligand, and $Pd_2(dba)_3$ were all indispensable for this transformation. Without any of them, *rac*-**1** was fully recovered as a racemate (entries 11−13). The base (K_2CO_3) was beneficial to enantioselectivity (entry 14).

Experimental Mechanistic Investigations. Several kinetic experiments and spectral studies were performed to probe the possible mechanism of this photochemical deracemization process (Figure 2). The yields and ee values of the model reaction was monitored first (Figure 2a). Rapid deracemization was observed, and the value of ee reached its peak in 2 h. At this period of time, decomposition of the allene was not observed. However, with the extension of the reaction time, racemization and decomposition of the enantioenriched allene occurred. The deracemization progresses of optically pure *R*-**1** and *S*-**1** under standard conditions were also monitored. The complete inversion of *R*-**1** to *S*-**1** was observed in 2.5 h (Figure 2b). *S*-**1** remained almost unchanged for 4 h in terms of the yields and optical purity (Figure 2c). It decomposed and racemized slowly with the extension of the reaction time. Ultraviolet−visible (UV−vis) spectroscopy of several species and emission spectra of light sources are shown in Figure 2d. The allene *rac*-**1** itself did not exhibit absorption in the visible region. The Pd(0) complex exhibited significant

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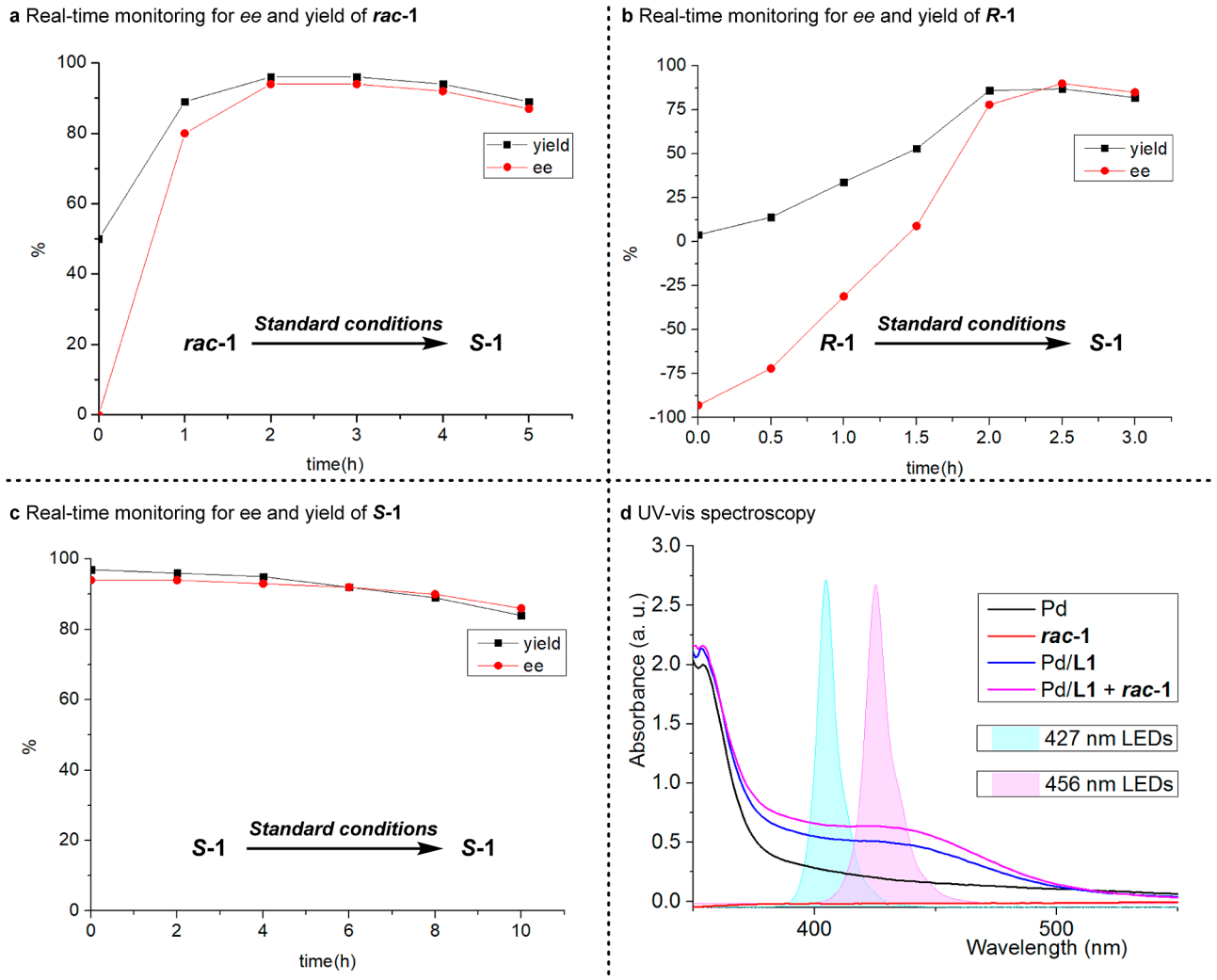


Figure 2. Experimental mechanistic studies of photochemical deracemization. (a) Real-time monitoring for ee and yield of *rac*-**1**. (b) Real-time monitoring for ee and yield of *R*-1. (c) Real-time monitoring for ee and yield of *S*-1. (d) UV−vis spectroscopy of the catalyst and *rac*-1 and emission spectra of light sources.

absorption in the range of 400−450 nm, and the addition of the chiral ligand **L1** and allene *rac*-**1** could enhance this absorption.

Proposed Mechanism and DFT Calculations. Based on the aforementioned kinetic and spectral data, a possible principle for this photochemical deracemization was proposed. As shown in Figure 3a, the coordination of Pd/**L1** with *R*-allene is favored due to ligand−substrate steric repulsions and the thermodynamically favored Pd/**L1**/*R*-allene complex is easier to be excited to undergo the metal-to-ligand charge transfer process.[71−74] Thus, the *R*-allene can racemize rapidly, while *S*-allene remains largely intact during the period of reaction time, which is the decisive factor for excellent deracemization efficiency.

We then conducted quantum mechanical calculations to verify this proposed reaction principle and to rationalize the asymmetric induction. To establish the basis, we commenced with the η^2-coordination of allene and the photoexcitation thereafter. As presented in Figure 3b, the ground-state **Pd-R**$_{S0}$ complex lies 2.2 kcal/mol below **Pd−S**$_{S0}$ due to the ligand−substrate steric repulsion in the latter species. The binding constants for *S*-allene or *R*-allene to the palladium complex Pd_2dba_3/**L1** were determined by ^{19}F NMR titration, and the fact that *R*-allene *R*-**49** binds stronger to the Pd complex than *S*-allene *S*-**49** [K_a (*R*-**49**) = 163.0 and K_a (*S*-**49**) = 22.7, respectively] is consistent with the density functional theory (DFT) calculations (for details, see Section 9.7 in the Supporting Information). The significant energy difference translates into stereoselective complexation of *R*-allene, which can lead to an accumulation of **Pd-R**$_{S0}$ over **Pd−S**$_{S0}$ and therefore amplify the consumption of the *R*-allene. Moreover, **Pd-R**$_{S0}$ displays a lower vertical excitation energy (E_{ex} = 71.9 kcal/mol for **Pd-R**$_{S0}$ vs 80.7 kcal/mol for **Pd−S**$_{S0}$). This lower E_{ex} of **Pd-R**$_{S0}$ translates to an absorption wavelength of 398 nm, which better matches our visible-light condition than the 355 nm for **Pd−S**$_{S0}$. Combined, these results implicate the easier generation of excited species for the Pd−*R*-allene complex.

We next turned to investigating the photochemical deracemization. Unlike the ground-state η^2-coordination, the photosensitized **Pd-R**$_{T1}$ and **Pd−S**$_{T1}$ feature σ-type Pd−C2 bonding, leaving an unpaired electron delocalized over C1 and C3. Despite the kinetic feasibility of isomerization via the low-barrier transition state **TS**$_{ISO,T1}$ (2.4 kcal/mol), the aforementioned bonding mode implies the rotation of the allyl C1═C2 bond along with its Np moiety, and the resultant

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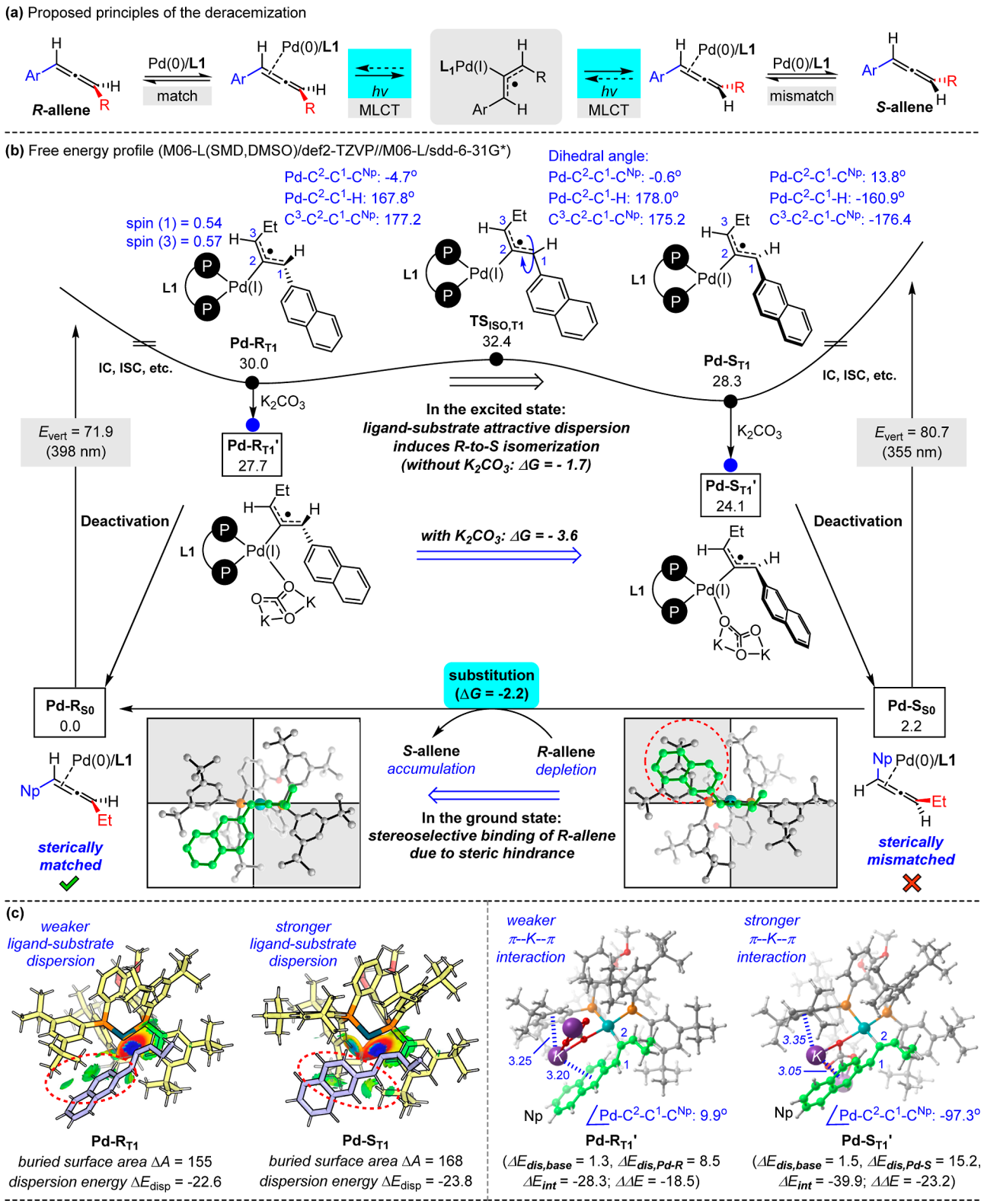


Figure 3. Proposed mechanism and its description of photochemical deracemization based on DFT studies. (a) Proposed principle of the deracemization. (b) Free energy profile. (c) Optimized structures with noncovalent interaction analysis and distortion/interaction analysis (the total distortion energy is divided into the distortion energies of the K_2CO_3 fragment and the **Pd-R**$_{T1}$/**Pd−S**$_{T1}$ fragment). Energies are given in kcal/mol, and areas are given in Å^2. The labeled bond lengths represent the distance from K to the center of the ring (unit: Å).

flexibility of relative substrate−ligand positions poses a challenge for the stereocontrol. Notably, our modeling unveiled a reversal of *R*−*S* relative energies on the T_1 manifold, with the targeted **Pd−S**$_{T1}$ now lying lower than **Pd-R**$_{T1}$ by 1.7 kcal/mol, leading to the stereoinduction. Besides, such a driving force is examined to result from stronger substrate−ligand attractive dispersion in **Pd−S**$_{T1}$ (Figure 3c), which comprises a distinct origin of stereo-

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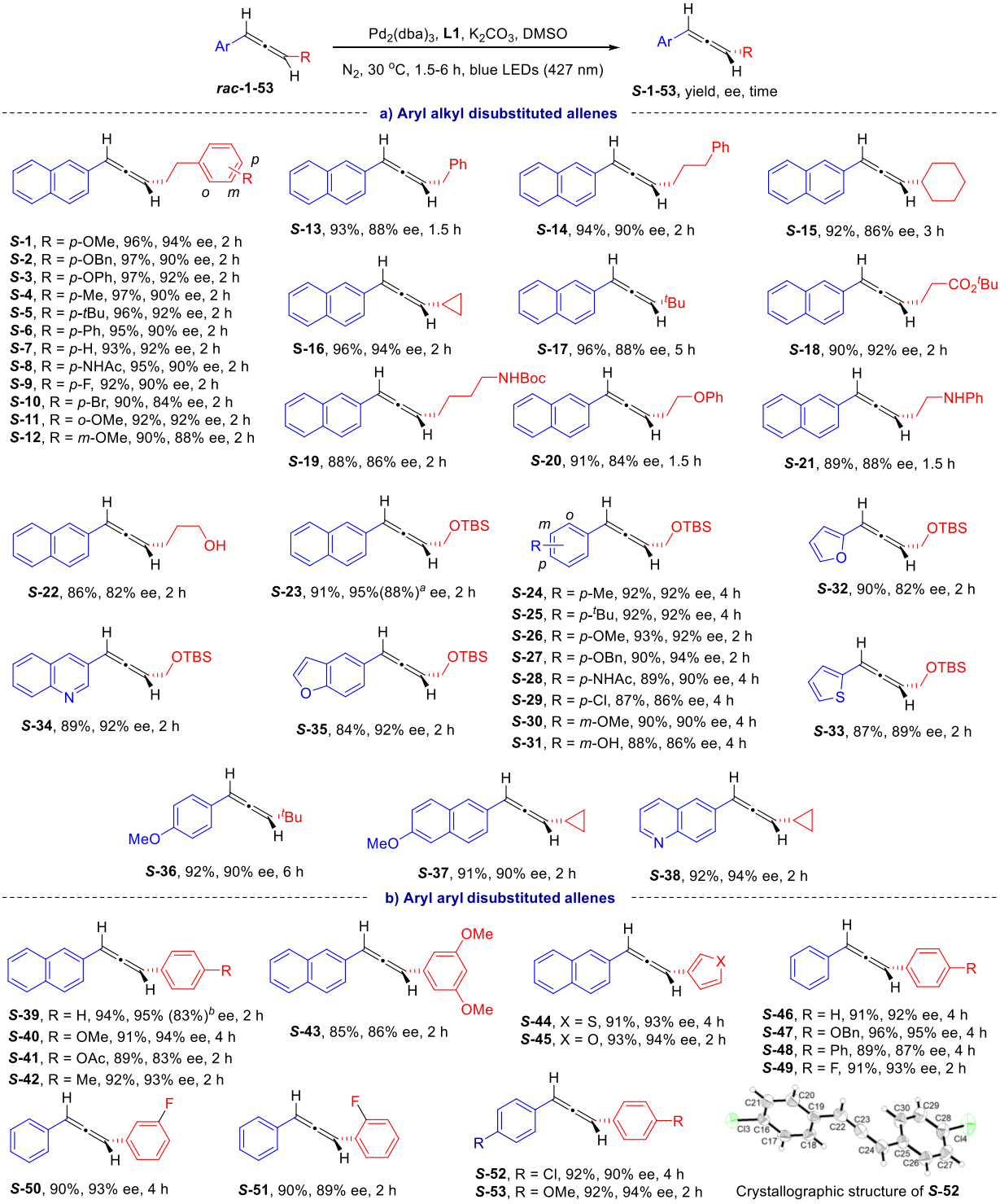


Figure 4. Substrate scope for the photochemical deracemization of allenes. Conditions: reactions were carried out with *rac*-**1** (0.1 mmol), $Pd_2(dba)_3$ (2.3 mg, 2.5 mol %), **L1** (2.7 mg, 6 mol %), and K_2CO_3 (0.15 mmol) in DMSO (2.0 mL) at 30 °C under 40 W blue LEDs (427 nm). The ee values were determined by HPLC on a chiral stationary phase. Isolated yields are the overall yields of *R*- and *S*-allenes as determined by flash column chromatography on silica gel. [a]Without K_2CO_3, 4 h; [b]Without K_2CO_3, 2 h.

differentiation compared with **Pd**-$\mathbf{R}_{S0}$ vs **Pd**−$\mathbf{S}_{S0}$.[75] Additionally, although it is difficult to identify the exact function of K_2CO_3, the significant base effect that enhances enantioinduction was revealed preliminarily from our calculations. Upon coordination to the Pd(I) center, K_2CO_3 could be sandwiched at the interspace between the ligand and substrate (Figure 3b/c). We propose that **Pd**−S'_{T1} benefits from a more favorable and well-aligned $\pi\cdots K\cdots\pi$ interaction compared to **Pd**-$\mathbf{R}'_{T1}$, which is supported by structural detection (dihedral angle of Pd−C^1−C^2−C^{Np}) and distortion/interaction analysis (Figure 3c).[76] These differential interactions can further amplify the energetic difference of the excited states, thereby improving the overall stereoselectivity and elucidating the important role of K_2CO_3 qualitatively. Hence, in addition to the reported

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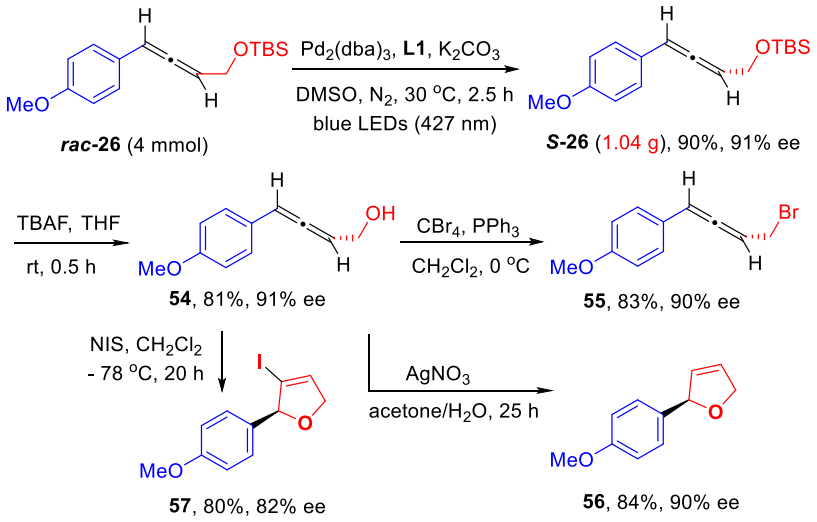


Figure 5. Gram-scale preparation and synthetic applications. TBAF = tetrabutyl ammonium fluoride. NIS = N-iodosuccinimide.

sensitization-rate differentiation, our new photoexcited Pd catalytic system can manipulate the energetic landscape to selectively recruit the *R*-allene at the ground state and effectively induces the *R*-to-*S* isomerization at the excited state, thus realizing asymmetric induction through a synergy of multiple contributors.

Substrate Scope and Limitation. With the optimal reaction conditions established, the generality and limitation of this deracemization protocol were evaluated (Figure 4). The cautious monitoring of the ee with time exhibited that the photochemical deracemization of different substrates reached photostationary states within 1.5−6 h under optimized conditions. To our delight, various aryl-alkyl disubstituted allenes were well-tolerated in this transformation and provided highly enantioenriched allenes in 89−97% yields with 86−95% ee. We first focused on the alkyl moieties. Substituents with different electronic properties at different positions on the phenyl rings of the alkyl chain had a negligible effect on the enantioselectivity (*S*-**1**−**12**). Chain length of the alkyl group and cyclic alkyl group (*S*-**13**−**17**) did not affect the enantioselectivity either. The sterically hindered *tert*-butyl group had no significant influence on this deracemization transformation (95% yield and 88% ee for *S*-17) in terms of yield and enantioselectivity, but with a longer reaction (6 h). A broad range of functional groups on the skeleton of aliphatic substituents, such as ester (*S*-**18**), amide (*S*-**19**), ether (*S*-**20**), amine (*S*-**21**), hydroxy group (*S*-**22**), and silyl ether (*S*-**23**), were all tolerated. To further expand the scope of the photochemical deracemization, a range of allenes with diverse substitution patterns on aryl moieties were then tested. Similarly, no obvious substitution effect was found (*S*-**24**−**31**). Heteroaromatic groups, such as furan (*S*-**32**), thiophene (*S*-**33**), quinoline (*S*-**34**), and benzofuran (*S*-**35**), proved to be competent reaction partners. Other structurally diverse allenes (*S*-**36**, *S*-**37**, and *S*-**38**) could also be prepared with good yields (91−92% yields) and enantioselectivities (80−94% ee). We next turned our attention to aryl−aryl disubstituted allenes. We were pleased to find that photochemical deracemization of aryl−aryl disubstituted allenes was also efficient, giving the desired chiral deracemized products (*S*-**39**−**53**) with excellent enantiopurity (86−95% ee) and yields (85−96%). Substituents with different electronic properties at different positions on the aryl rings of aryl−aryl disubstituted allenes were well tolerated in this reaction. Allenes containing thiophene and furan (*S*-**44** and *S*-**45**) proved to be competent reaction partners. Disappointedly, alkyl−alkyl disubstituted and trisubstituted allenes tested so far were not effective under standard conditions in terms of enantioselectivity (For more unsuccessful substrates, see Figure S1 in the Supporting Information). The absolute configuration of products was established ambiguously by single-crystal X-ray diffraction analysis of allene *S*-**52**.[77]

Synthetic Practicality and Applications. To demonstrate the synthetic utility of this protocol, a scale-up reaction and some useful transformations were conducted (Figure 5). The photochemical deracemization of *rac*-**26** on a 4.0 mol scale was applied under the optimal reaction conditions. This scaled reaction worked smoothly, and the enantioenriched product *S*-**26** (1.04 g) was obtained in 4 h with a comparable yield and enantioselectivity (90% yield and 91% ee). The TBS group of deracemization product *S*-**26** could be conveniently removed with TBAF in THF to provide synthetically versatile allenyl alcohol **54** without loss of any optical purity. Subsequently, a series of transformations from allenyl alcohol **54** were performed. The allenyl alcohol **54** could be converted to the corresponding allenyl bromide **55** under Appel's conditions (PPh_3 and CBr_4) in an 83% yield without affecting the optical purity (90% ee).[78] The enantioenriched allenyl alcohols were identified as valuable intermediates in the construction of chiral heterocycles. Treatment of the allenyl alcohol **54** with $AgNO_3$ in aqueous acetone solution provided a dihydrofuran **56** (84% yield and 90% ee) with complete axial-to-point chirality transfer.[79] An iodinated dihydrofuran **57** was also accessed in the assistance of NIS in an 80% yield with slight loss of optical purity (82% ee).[80]

CONCLUSIONS

In summary, we have developed an unprecedented photoexcited palladium-catalyzed deracemization of disubstituted allenes under mild reaction conditions. A series of structurally diverse disubstituted allenes bearing primary alcohol, sensitive secondary amine, ester, N-Boc, ether, etc. were suitable in the established protocol with excellent yields and enantioselectivities. The resultant functionalized enantioenriched allenes can be easily transformed to a series of synthetically useful

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molecules. The employment of a photoexcited chiral palladium complex is crucial to the success of this protocol, which could not be achieved under thermal activation. Mechanism studies implicate dual modes of asymmetric induction: (1) sterically controlled stereoselective binding of one allene enantiomer under the ground-state and (2) facile, noncovalent interaction-driven excited-state isomerization toward the opposite enantiomer. The success of this newly established photochemical deracemization strategy should provide inspiration for expansion to other multisubstituted allenes and will open up a new mode for enantioselective excited-state palladium catalysis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c07126.

General experimental procedures and characterization and ^{1}H NMR, ^{13}C NMR, and ^{19}F NMR spectral data (PDF)

Accession Codes

CCDC 2290491 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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